

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.12758	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	76.89182	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1910.16218	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	1.0	0.074405	0.074405	255.19108	2	0	0	0.937227	14.955
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	248.82992	2	0	0	0.961677	14.14
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	142.37444	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.12758.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 76.89182.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1910.16218.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.079232.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.074405.
- Held-out runtime ms: 255.19108.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.001657.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.099452, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.09508, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.067432, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.052353, errors=[]

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 248.82992.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 142.37444.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Aspect	Summary
Direct Synthesis	phase9_closeout_direct_generate_plus_one_repair achieved 100% feasibility with a 0.0744 feasible gap on heldout, close to best baseline, moderate runtime (~255 ms), and high novelty (~0.94). Complexity moderate (~15).
Budget-Matched Independent Search	phase9_closeout_budget_matched_no_replay failed feasibility completely (0%), penalized gap fixed at 3.0, despite similar runtime to direct synthesis (~249 ms), high novelty (0.96), moderate complexity (~14). No useful solutions found.
Replay-Aware Iterative Search	phase9_closeout_replay_solver_evolution had full feasibility (100%) but a large feasible gap (0.273), runtime ~142 ms, zero novelty and complexity, effectively matching nearest neighbor baseline in quality but worse than Clarke-Wright. Demonstrated no gap improvement vs no-replay.
Fixed-Baseline Context	Baselines showed consistent full feasibility. Clarke-Wright savings baseline was best with lowest heldout feasible gap (0.0738) but longer runtime (~77 ms). Nearest neighbor had higher gap (0.273), faster runtime (~42 ms), and regret insertion local search was slow (~1910 ms) with worst gaps (~0.466).
Feasibility	All except budget-matched no-replay found 100% feasible solutions on heldout.

Aspect	Summary
Objective Gap	Best gap: Clarke-Wright (0.0738), direct synthesis close (0.0744), others notably worse.
Runtime	Nearest neighbor fastest (~42 ms), Clarke-Wright (~77 ms), replay solver (~142 ms), direct synthesis (~255 ms), budget-matched no-replay (~249 ms), regret insertion very slow (~1910 ms).
Robustness (Dispersion)	Low dispersion (best robustness) in direct synthesis (≈ 0.0017), slightly higher for Clarke-Wright (≈ 0.0078) and regret insertion.
Learned Condition vs Fixed Baselines	No learned approach clearly outperformed the best fixed baseline (Clarke-Wright). Direct generate + repair matched gap closely but was slower.
Replay vs No-Replay	Replay-aware search improved feasibility and reduced gap compared to no-replay search (which failed feasibility), but did not reach best baseline quality or outperform direct synthesis.

Key Conclusions

- Clarke-Wright savings baseline remains top performer in objective gap and feasibility with moderate runtime.
- Direct generate plus one repair learned method closely matches Clarke-Wright's objective gap with perfect feasibility, albeit slower and more complex.
- Budget-matched no-replay search failed feasibility and produced invalid solutions.
- Replay-aware solver improves over no-replay in feasibility and quality but does not exceed fixed baseline or direct synthesis.
- No learned or replay method clearly beats Clarke-Wright baseline in gap or combined metrics under this evaluation.